

Good semen quality and life expectancy: a cohort study of 43,277 men.

Fertility status may predict later mortality, but no studies have examined the effect of semen quality on subsequent mortality. Men referred to the Copenhagen Sperm Analysis Laboratory by general practitioners and urologists from 1963 to 2001 were, through a unique personal identification number, linked to the Danish central registers that hold information on all cases of cancer, causes of death, and number of children in the Danish population. The men were followed until December 31, 2001, death, or censoring, whichever occurred first, and the total mortality and cause-specific mortality of the cohort were compared with those of all age-standardized Danish men or according to semen characteristics. Among 43,277 men without azospermia referred for infertility problems, mortality decreased as the sperm concentration increased up to a threshold of 40 million/mL. As the percentages of motile and morphologically normal spermatozoa and semen volume increased, mortality decreased in a dose-response manner ($P(\text{trend}) < 0.05$). The decrease in mortality among men with good semen quality was due to a decrease in a wide range of diseases and was found among men both with and without children; therefore, the decrease in mortality could not be attributed solely to lifestyle and/or social factors. Semen quality may therefore be a fundamental biomarker of overall male health.